

السلام عليكم الله ورحمته وبركاته

الله "وَتُعْزَمَنْ تَشَاءُ وَتَذَلُّ مَنْ تَشَاءُ"

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Where Arise Clique Cover? (pg176)

Answer:

The clique cover problem arises in applications of clustering. We put an edge between two nodes if they are similar enough to be clustered in the same group. We want to know whether it is possible to cluster all the vertices into k groups.

Q:5 How to get Knapsack optimal solution with dynamic programming algorithm table? (5)

Answer: (Page 96)

The algorithm for computing $V[I, j]$ does not keep record of which subset of items gives the optimal solution. To compute the actual subset, we can add an auxiliary Boolean array $keep[I, j]$ which is 1 if we decide to take the i th item and 0 otherwise. We will use all the values $keep[I, j]$ to determine the optimal subset T of items to put in the knapsack as follows:

- If $keep[n, W]$ is 1, then $n \in T$. We can now repeat this argument for $keep[n - 1, W - w_n]$.
- If $keep[n, W]$ is 0, then $n \notin T$ and we repeat the argument for $keep[n - 1, W]$.

What are common problems related to Communication Network and Circuit designing.(2)

Answer:

A common problem is communications networks and circuit design is that of connecting together a set of nodes by a network of total minimum length. The length is the sum of lengths of connecting wires.

Diff between Forward And Cross Edge.(2)

Answer:

Forward edge: (u, v) where v is a proper descendent of u in the tree.

Back edge: (u, v) where v is an ancestor of u in the tree.

Cross edge: (u, v) where u and v are not ancestor or descendent of one another.

Write the computational Time of Bellman ford Algorithm.(2)

Answer:

Bellman-Ford applies relaxation to every edge of the graph and repeats this $V - 1$ times.

How Dijkstra's algorithm works. (3)

Answer:

Dijkstra's algorithm is a simple greedy algorithm for computing the single-source shortest-paths to all other vertices.

Dijkstra's algorithm works on a weighted directed graph $G = (V, E)$ in which all edge weights are non-negative, i.e., $w(u, v) \geq 0$ for each edge $(u, v) \in E$.

What is polynomial time Algorithms? Give Example (3)

Answer:

A polynomial time algorithm is any algorithm that runs in $O(n^k)$ time. A problem is solvable in polynomial time if there is a polynomial time algorithm for it

What Is Activity scheduling?? Give example (3)

Answer:

Activity Selection:

The activity scheduling is a simple scheduling problem for which the greedy algorithm approach provides an optimal

solution. We are given a set $S = \{a_1, a_2, \dots, a_n\}$ of n activities that are to be scheduled to use some resource. Each activity a_i must be started at a given start time s_i and ends at a given finish time f_i .

An example is that a number of lectures are to be given in a single lecture hall. The start and end times have been set up in advance. The lectures are to be scheduled. There is only one resource (e.g., lecture hall)

Why Reduction is used? Give an example (5) Most important

Answer:-

The class NP-complete problems consist of a set of decision problems that no one knows how to solve efficiently. But if there were a polynomial solution for even a single NP-complete problem, then every problem in NPC will be solvable in polynomial time. For this, we need the concept of reductions.

Using Prim's Algorithm Find Minimum Spanning Tree (5) Most important

Answer:

Prim's algorithm builds the Minimum Spanning Tree by adding leaves one at a time to the current tree. We start with a root vertex r ; it can be any vertex. We look to add a single vertex as a leaf to the tree.

What is minimizing spanning tree?

Answer:

A minimum spanning tree is a tree of minimum weight.

The computational problem is called the minimum spanning tree (MST) problem.

Differentiate between back edge and forward edge.

Answer: (Page 128)

Back edge: (u, v) where v is an ancestor of u in the tree.

Forward edge: (u, v) where v is a proper descendent of u in the tree.

Given a graph $G(V, E)$ any DFS forest of G and consider edge $(u, v) \in E$. prove that if this edge is tree, forward or back edge then $f[u] > f[v]$ and if this edge is cross edge then $f[u] \leq f[v]$

Answer: (Page 130)

Proof:

For the non-tree forward and back edges the proof follows directly from the parenthesis lemma. For example, for a forward edge (u, v) , v is a descendent of u and so v 's start-finish interval is contained within u 's implying that v has an earlier finish time. For a cross edge (u, v) we know that the two time intervals are disjoint. When we were processing u , v was not white (otherwise (u, v) would be a tree edge), implying that v was started before u . Because the intervals are disjoint, v must have also finished before u .

Consider a graph $G=(V, E)$ and any DFS forest for G , prove that G has a cycle if and only if DFS forest has a back-edge?(5 marks)

Answer:

Proof: If there is a back edge (u, v) then v is an ancestor of u and by following tree edge from v to u , we get a cycle.

We show the contrapositive: suppose there are no back edges. By the lemma above, each of the remaining types of edges, tree, forward, and cross all have the property that they go from vertices with higher finishing time to vertices with lower finishing time. Thus along any path, finish times decrease monotonically, implying there can be no cycle.

0---->1, 1---->4, 4--->0, 0----->2, 2---->3, 3--->2 list the strong components

Suppose you could prove that an NP-complete problem can be solved in polynomial time. What would be the consequence?

Answer: Page 173

If we can solve a problem in polynomial time, we can certainly verify the solution in polynomial time.

More formally, we do not need to see a certificate to solve the problem; we can solve it in polynomial time anyway.

However, it is not known whether $P = NP$. It seems unreasonable to think that this should be so. Being able to verify that you have a correct solution does not help you in finding the actual solution. The belief is that

$P \neq NP$ but no one has a proof for this.

Define Cross Edge? (2 marks) repeat

Cross Edge : (u, v) in this u and v is not decendent or ancestor to one with eachother

Forward edge (u, v) where u is a ancestor of u in the tree.

Define Free Tree? (2 marks)

Answer: (Page 142)

A free tree is a tree with no vertex designated as the root vertex.

What are out-degree and in-degree of a vertex in graphs? (2 marks)

Answer:

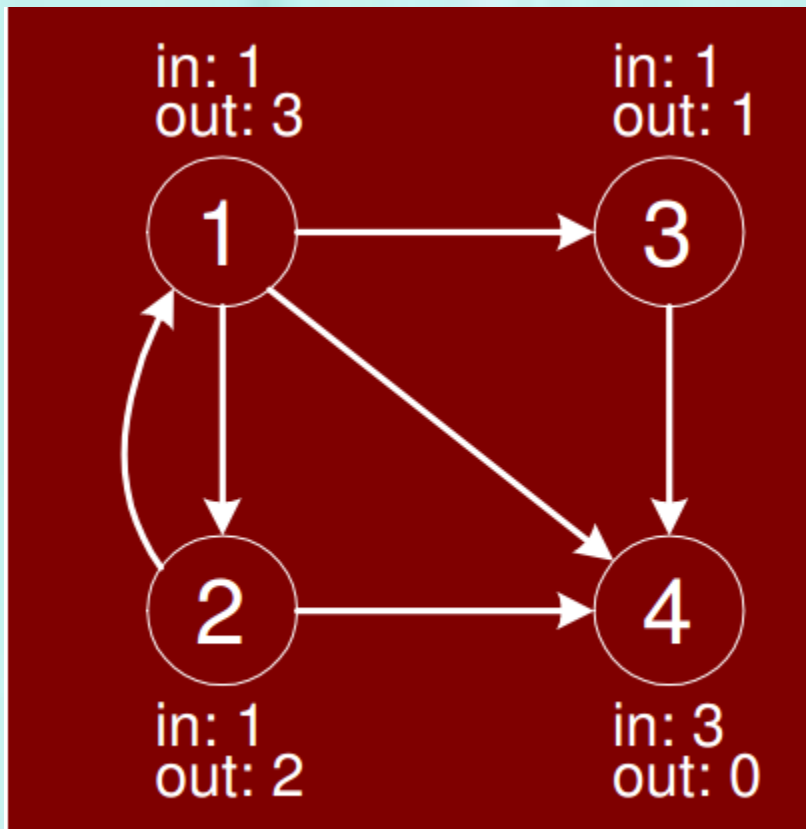


Figure 8.4: In and out degrees of vertices of a graph

What is Bellman-Ford running time? (2 marks)

Answer:

The algorithm is slower than Dijkstra's, running in $\Theta(VE)$ time.

Give a pseudo code for counting money problem. Take largest note or coin first?(3 marks)

Answer:

Suppose you want to count out a certain amount of money, using the fewest possible bills and coins. A greedy algorithm to do this would be: at each step, take the largest possible note or coin that does not overshoot.

```
while (N > 0) {  
  give largest denomination coin  $\leq$  N  
  reduce N by value of that coin  
}
```

Consider the currency in U.S.A. There are paper notes for one dollar, five dollars, ten dollars, twenty dollars, fifty dollars and hundred dollars. The notes are also called “bills”. The coins are one cent, five cents (called a “nickle”), ten cents (called a “dime”) and twenty five cents (a “quarter”).

Describe three variants of shortest path problem?(3 marks)

There are a few variants of the shortest path problem.

Single-source shortest-path problem: Find shortest paths from a given (single) source vertex $s \in V$ to every other vertex $v \in V$ in the graph G .

Single-destination shortest-paths problem: Find a shortest path to a given destination vertex t from each vertex v . We can reduce this problem to a single-source problem by reversing the direction of each edge in the graph.

Single-pair shortest-path problem: Find a shortest path from u to v for given vertices u and v . If we solve the single-source

problem with source vertex u , we solve this problem also. No algorithms for this problem are known to run asymptotically faster than the best single-source algorithms in the worst case. All-pairs shortest-paths problem: Find a shortest path from u to v for every pair of vertices u and v . Although this problem can be solved by running a single-source algorithm once from each vertex, it can usually be solved faster.

Ref: Handouts Page No.153

According to Dijkstra's Algorithm, write the pseudo code to relax a vertex?(5 marks)

Answer:

RELAX((u, v))

1 if $(d[u] + w(u, v) < d[v])$

2 then $d[v] \leftarrow d[u] + w(u, v)$

3 $pred[v] = u$

Over all time for Kruskal Algorithm if the Graph is sparse?

Answer:

Overall time for Kruskal is $\Theta(E \log E) = \Theta(E \log V)$ if the graph is sparse.

What are the two points that assume while providing the correctness of dijkstra algorithm?

Answer:

We will prove the correctness of Dijkstra's algorithm by Induction.

We will use the definition that $d(s, v)$ denotes the minimal distance from s to v .

For the base case

1. $S = \{s\}$
2. $d(s) = 0$, which is $d(s, s)$

Diagram given and tell the weight of MST 5 marks

Answer:

A common problem in communications networks and circuit design is that of connecting together a set of nodes by a network of total minimum length. The length is the sum of lengths of connecting wires.

Consider, for example, laying cable in a city for cable t.v.

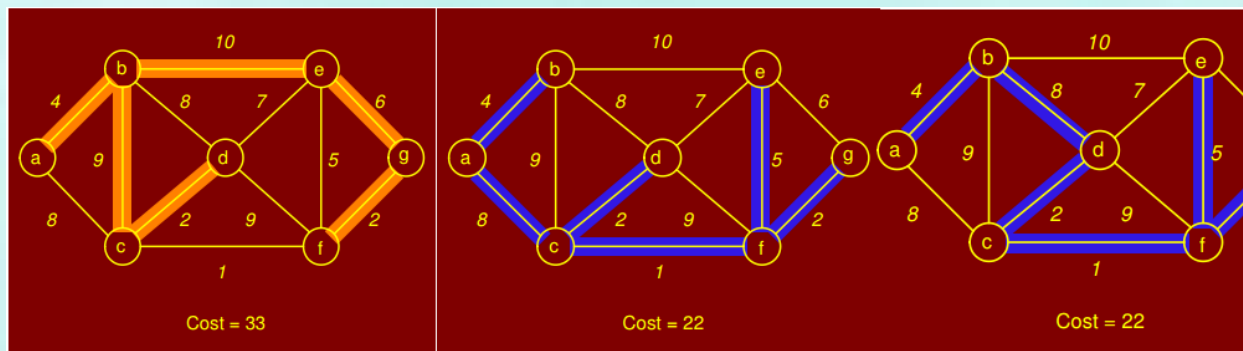
The computational problem is called the minimum spanning tree (MST) problem. Formally, we are given a connected, undirected graph $G = (V, E)$ Each edge (u, v) has numeric weight of cost.

We define the cost of a spanning tree T to be the sum of the costs of edges in the spanning tree

$$w(T) = \sum_{(u,v) \in T} w(u,v)$$

A minimum spanning tree is a tree of minimum weight.

Figures ??, ?? and ?? show three spanning trees for the same graph. The first is a spanning tree but is not a MST; the other two are.



Which algorithm best for Lahore to Islamabad track that use minimum link? 5 marks

What is free tree

Answer:

A free tree is a tree with no vertex designated as the root vertex.

What is cross edge

Answer:

Cross edge: (u, v) where u and v are not ancestor or descendent of one another.

What is running time of dijkstra algorithm

Answer:

Running time is the same as Prim's algorithm, i.e., $\Theta(E \log V)$.

What do you mean by polynomial time algorithm? Explain what kind of problems can be solved by using polynomial time algorithm?

Answer: (Page 169)

A polynomial time algorithm is any algorithm that runs in $O(n^k)$ time.

A problem is solvable in polynomial time if there is a polynomial time algorithm for it.

When Floyd warshall algorithm introduced

Answer:

The algorithm dates back to the early 60's.

Heap sort and heap order define krna tha (2 marks)

Answer:

- A heap is a left-complete binary tree that conforms to the heap order.
- The heap order property: in a (min) heap, the parent node has key smaller than or equal to both of its children nodes.
- Similarly, in a max heap, the parents have a key larger than or equal both of its children.

Recursive explanation of dynamic programming .

Formulate the problem into smaller sub problems, find optimal solution to these sub problems in a bottom up fashion then write an algorithm to find the solution of whole problem starting with base case and works it way up to final solution.

Steps in the dynamic programming strategy:

- 1. Simple Subproblems:** We should be able to break the original problem to smaller subproblems that have the same structure
- 2. Principle of Optimality:** Recursively define the value of an optimal solution. Express the solution of the original problem in terms of optimal solutions for smaller problems.
- 3. Bottom-up computation:** Compute the value of an optimal

solution in a bottom-up fashion by using a table structure.

4. Construction of optimal solution: Construct an optimal solution from computed information. Dynamic programming is essentially recursion without repetition. Developing a dynamic programming algorithm generally involves two separate steps:

Formulate problem recursively. Write down a formula for the whole problem as a simple combination of answers to smaller sub problems. Build solution to recurrence from bottom up. Write an algorithm that starts with base cases and works its way up to the final solution.

Ref: Handouts Page No.75

Give a detailed example for 2d maxima problem

Answer: -

The problem with the brute-force algorithm is that it uses no intelligence in pruning out decisions.

For example, once we know that a point p_i is dominated by another point p_j , we do not need to use p_i for eliminating other points. This follows from the fact that dominance relation is transitive. If p_j dominates p_i and p_i dominates p_h then p_j also dominates p_h ; p_i is not needed.

Ref: Handouts Page No.17

How the generic greedy algorithm operates in minimum spanning tree?

Answer:-

A generic greedy algorithm operates by repeatedly adding any safe edge to the current spanning tree. When is an edge safe?

Consider the theoretical issues behind determining whether an edge is safe or not.

Let S be a subset of vertices $S \subseteq V$. A cut $(S, V - S)$ is just a partition of vertices into two disjoint subsets. An edge (u, v) crosses the cut if one endpoint is in S and the other is in $V - S$. Given a subset of edges A , a cut respects A if no edge in A crosses the cut. It is not hard to see why respecting cuts are important to this problem. If we have computed a partial MST and we wish to know which edges can be added that do not induce a cycle in the current MST, any edge that crosses a respecting cut is a possible candidate.

Ref: Handouts Page No.143

What are two cases for computing

Answer:-

There are two cases for computing L_{ij} the match case if $a_i = b_j$, and the mismatch case if $a_i \neq b_j$. In the match case, we do the following:

Ref: Handouts Page No.143

Describe Minimum Spanning Trees Problem with examples.

Problem

Given a connected weighted undirected graph, design an algorithm that outputs a minimum spanning tree (MST) of

Examples

The graph is a complete, undirected graph $G = (V, E, W)$, where V is the set of pins, E is the set of all possible interconnections between the pairs of pins and $w(e)$ is the length of the wire needed to connect the pair of vertices

Each edge has a cost. The minimum spanning tree is a least total cost tree. For example:

Ref: Handouts Page No.142

What is decision problem, also explain with example?

A decision problem is a question in some formal system. A problem is called a decision problem if its output is a simple “yes” or “no” (or you may phrase this as true/false, 0/1, accept/reject.) We will phrase many optimization problems as decision problems.

For example, the MST decision problem would be: Given a weighted graph G and an integer k , does G have a spanning tree whose weight is at most k ?

Ref: Handouts Page No.170

Do you think greedy algorithm gives an optimal solution to the activity scheduling problem?

The greedy algorithm gives an optimal solution to the activity scheduling problem.

Proof:

The proof is by induction on the number of activities. For the basis case, if there are no activities, then the greedy algorithm is trivially optimal. For the induction step, let us assume that the greedy algorithm is optimal on any set of activities of size strictly smaller than $|S|$ and we prove the result for S . Let S_0 be the set of activities that do not interfere with activity a_1 . That is

Any solution for S_0 can be made into a solution for S by simply adding activity a_1 , and vice versa. Activity a_1 is in the optimal schedule (by the above previous claim). It follows that to produce an optimal schedule for the overall problem, we should first schedule a_1 and then append the optimal schedule for S_0 . But by induction (since $|S_0| < |S|$), this exactly what the greedy algorithm does.

Ref: Handouts Page No.109

Define Forward edge

The most natural result of a depth first search of a graph (if it is considered as a function rather than procedure) is a spanning tree of the vertices reached during the search. Based on this spanning tree, the edges of the original graph can be divided into three classes: forward edges (or "discovery edges"), which point from a node of the tree to one of its descendants,

Ref: Handouts Page No.129 130

Is there any relationship between number of back edges and number of cycles in DFS?

Answer:

The time stamps given by DFS allow us to determine a number of things about a graph or digraph. we can determine whether the graph contains any cycles.

Lemma: Given a digraph $G = (V, E)$, consider any DFS forest of G and consider any edge $(u, v) \in E$. If this edge is a tree, forward or cross edge, then $f[u] > f[v]$. If this edge is a back edge, then $f[u] \leq f[v]$.

Proof: For the non-tree forward and back edges the proof follows directly from the parenthesis lemma. For example, for a forward edge (u, v) , v is a descendent of u and so v 's start-finish interval is contained within u 's implying that v has an earlier finish time. For a cross edge (u, v) we know that the two time intervals are disjoint. When we were processing u , v was not white (otherwise (u, v) would be a tree edge), implying that v was started before u . Because the intervals are disjoint, v must have also finished before u .

Ref: Handouts Page No.130

What is the difference between directed graph and undirected graph?

Ans:

Directed graph is representation of edges with explicit directions from vertices whereas undirected graphs have no explicit arrows to traverse; rather we can traverse both ways from one vertex to other.

Edit distance matrix

First of all focus on the recurrence equation in which you can find the corresponding values based on variables " i " and " j " the movements in rows, columns and diagonally based on constraints

.In table the horizontal arrow is for "INSERTION"

Downward vertical arrow is for "DELETION"

diagonal arrow has meaning of “SUBSTITUTION” and “MAINTAIN” whichever is applicable.

Spanning tree

Well dear student spanning tree concept is simple; a tree having all the nodes of the graph connected generating no cycle. And time stamp concept is relating to the traversing when first time you have visited the node place some count there so you can reuse that thing for your purpose i.e. here Depth First Search (DFS) time stamped version can help you to understand all the concept just focus on the discussed example in handout

Provide By Vu Toper RM